

The speed of sound in a two-dimensional hard-disk gas: a divider-mode measurement against the equation of state

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Abstract—We measure the adiabatic speed of sound of a hard-disk fluid from the eigenfrequency of a massive divider separating two compartments, the method of Román *et al.*, over 35 packing fractions from $\eta = 0.0065$ to 0.76 and nine divider masses per density. The measurement is compared with the Kolafa-Rottner equation of state through the adiabatic relation $c_s^2 = (k_B T/m)[Z + \eta Z' + Z^2]$ with no fitted parameter. For $\eta \leq 0.4$ the agreement is +1.01 % on average at $N = 100$; a size ladder to $N = 2500$ shows the offset is finite size. We bound the estimator’s own systematic below 0.3 %, and we show that the apparent density-dependent excess reported in 2002 is an artefact of the $\gamma = 2$ mapping used then: re-mapped adiabatically, that 24-year-old data set shows the same flat +1.1 % offset as ours, in the same box.

Index Terms—hard-disk fluid, speed of sound, event-driven molecular dynamics, equation of state, finite-size scaling

I. INTRODUCTION

THE hard-disk fluid is the simplest system with a non-trivial equation of state, and its thermodynamics is known to high precision from simulation [1]. Its *dynamical* response is measured less often. The speed of sound is the natural bridge: for a two-dimensional monatomic fluid it is fixed entirely by the equation of state,

$$c_s^2 = \frac{k_B T}{m} [Z + \eta Z' + Z^2], \quad (1)$$

so a measurement of c_s tests $Z(\eta)$ with the equation of state as an *input* rather than a fit. Agreement is then a check; disagreement is a measurement of something the equation of state does not contain.

Measuring c_s by launching a pulse needs a large box and care with the pulse amplitude. Román *et al.* [3] avoid both by placing a massive divider between two compartments and watching it ring: the divider is a mass on two gas springs and its eigenfrequency is set by c_s . The measurement becomes a frequency, which is cheap to make precise, and the system can be small. We adopt that apparatus, and much of this paper is about what it takes to trust the frequency it returns.

Two things motivate revisiting it. First, the estimator. The divider carries a second, much slower motion that is not the sound mode, whose spectral weight can exceed the resonance;



Fig. 1. The apparatus, drawn by the simulation itself. Two compartments of 50 disks either side of the divider. Walls are labelled by mass; a held wall is drawn solid and a free one hollow. The wall at left bounds an inert compartment belonging to a companion experiment and is held immovable throughout this work.

an ungarded peak-finder is wrong by tens of per cent (§III). Second, the comparison. Román *et al.* convert Z to c_s with a constant $\gamma = 2$ and report an excess over theory that grows with density; we show in §VII that the growth is the mapping, and that their data and ours then agree.

II. MODEL AND APPARATUS

Hard disks of diameter $\sigma = 2r$ and mass m occupy a box of height H split at the centre by a divider of mass M and thickness t , with N_s disks on each side. Units are reduced, $k_B T = m = \sigma = 1$; we use $H = 10$ and $N_s = 50$ unless stated.

The dynamics are event-driven and exact between collisions; all results use the scientific backend. The divider is held for an equilibration period, during which each compartment is separately equalised to $KE = N_s k_B T$, and is then released; its position $x(t)$ is recorded at fixed sampling interval. Every trajectory is checked against a health contract — a scheduler forced advance, an overlap repair, a wall-position clamp or an overdue wall collision each discard that trajectory, never the cell. Of 7875 trajectories in the main campaign, 2 were discarded.

a) *Effective length*: The disk centres in one compartment span r to $L_0 - t/2 - r$, so the length entering the mode is

$$L_{\text{eff}} = L_0 - 2r - t/2. \quad (2)$$

Carrying t lowers every c_s by $(t/2)/(L_0 - 2r)$: 0.004 % at $\eta = 0.0065$ and 0.50 % at $\eta = 0.65$. The density is defined on the physical area, $\eta = N_s \pi r^2 / (L_0 H)$, with no exclusion — L_{eff} is an acoustic length and enters nowhere else.

b) *Integer-pixel box width*: The simulation sizes its box in whole pixels of $\sigma/24$, so when $2L_0$ is not a whole number of pixels the box is shorter than requested by $\delta = (48L_0 - \lfloor 48L_0 \rfloor)/24$, less than one pixel. The divider starts L_0 from

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one wall and oscillates about the centre of the shortened box, so each compartment is on average $L_0 - \delta/2$ long, as the recorded box centre confirms at every $N = 100$ density to $5.5 \times 10^{-6} \sigma$. For every density and system size we therefore use $\eta = N_s \pi r^2 / [(L_0 - \delta/2)H]$ and $L_{\text{eff}} = L_0 - 2r - t/2 - \delta/2$; $\delta = 0$ at 7 of the 35 densities, and the correction lowers c_s by at most 0.09 % for $\eta \leq 0.16$ and 0.47 % overall and raises η by at most 0.38 %. The correction was applied after the measurements, under a rule fixed before it was evaluated: the table is regenerated if any density's deviation from Kolafa-Rottner changes by more than half that density's error bar, which happened at six densities, all of which moved toward the equation of state. The same rule, pre-registered separately, was applied to the larger systems of §VI: the correction moves no point of the size ladder by more than 0.73 of its error bar and no fitted c_∞ by more than 0.34 of its error, but it lowers the coefficient of $1/\sqrt{N}$ by up to 1.10 of its error (at $\eta = 0.50$ and 0.65), so the size ladder was regenerated too.

III. THE ESTIMATOR

For each trajectory we form the periodogram of the divider position,

$$P(f) = \left| \sum_j (x(t_j) - \bar{x}) e^{-2\pi i f t_j} \right|^2, \quad (3)$$

and take the largest bin at or above a one-sided floor,

$$\nu = \arg \max_{f \geq \nu_{\text{pred}}/X} P(f), \quad X = 2.5, \quad (4)$$

then average over 25 seeds. With record length T the bin width is $1/T$, so at 200 predicted periods the rounding floor on one frequency is $100/(N_{\text{cyc}}\sqrt{12}) = 0.144$ %.

a) *Why a floor is needed:* Besides the resonance the divider performs a slow wander: a five-period running mean of x carries 9–57 % of the variance of x . It is not a start-up transient — its variance is the same in the second half of a record as the first, and a hundredfold longer hold does not change it — and it is a relaxation rather than an oscillation, with a correlation time of 53–102 acoustic periods. It is the adiabatic-piston mechanism: thermal fluctuations of the two compartment temperatures shift the pressure balance, the divider follows, and it returns only as heat crosses the divider. A dedicated run logging the compartment kinetic energies confirms it directly, with correlation $r = +0.978$ between the slow mode and the temperature difference and the amplitude reproduced to 6–7 % by $x = (L_0/2)(\Delta T/T) Z/(Z + \eta Z')$. Its power sits at low frequency, and at $\eta \geq 0.52$ it reaches 85–280 % of the resonance peak for the lightest divider.

The consequence is quantitative. Against a reference that windows the spectrum around the resonance, an unguarded largest bin is wrong by up to 33 % on 25-period records and up to 93 % at 200 periods — the error grows with record length, because a longer record resolves the wander into more bins, any of which can outgrow the resonance. Short records hide the problem rather than remove it. The floor removes it, and the answer does not depend on where the floor is placed: every X from 2.08 to 6.25 gives the identical c_s at 200 periods.

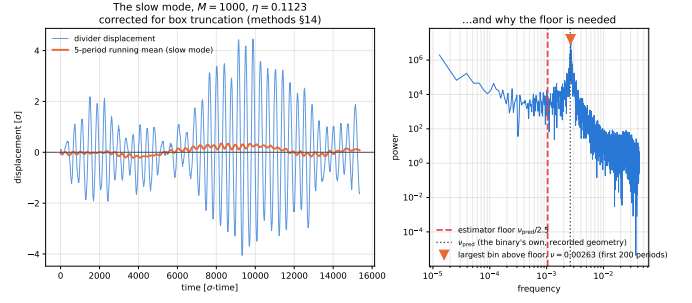


Fig. 2. Why the floor exists. **Left:** the divider displacement at $M = 1000$, $\eta = 0.1122$, with its five-period running mean — the slow adiabatic-piston wander, which carries no information about c_s but has more low-frequency power than the resonance. **Right:** the same record's periodogram. Without a floor the largest bin is the wander, not the mode; the estimator therefore takes the largest bin at $f \geq \nu_{\text{pred}}/2.5$.

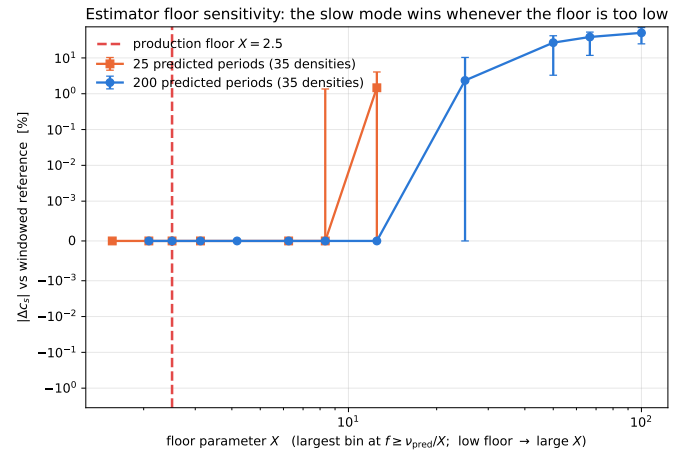


Fig. 3. Floor sensitivity. Median $|\Delta c_s|$ against the windowed reference over all 35 densities, with the 16–84 % spread as the error bar, versus the floor parameter X (the floor is ν_{pred}/X , so a large X is a low floor). The collapse at large X is the slow mode of Fig. 2 winning the periodogram; it is worse at 200 periods, where the wander has more bins below the resonance. At the production value $X = 2.5$ the estimator reproduces the windowed reference exactly at both record lengths.

b) *Largest bin or a fit?:* A resonance fit to each trajectory is the obvious alternative and the more precise one. Both put the nine masses of §IV on a single line to 0.27 %. But a biased estimator does not scatter about the line, it leans: regressing the residual on $\ln \alpha$ gives -0.286 ± 0.020 % per e-fold for the fit (14.4σ) against -0.030 ± 0.029 % for the largest bin (1.0σ , flat). The fit walks monotonically from +0.35 % at the lightest divider to -0.64 % at the heaviest. We use the largest bin and report the fit as a cross-check; no equation of state entered that choice.

c) *Bounding the rest:* Damping shifts the position-spectrum peak below the eigenfrequency by $\Gamma^2/4f_0^2$; with the measured $\Gamma/f_0 = 0.023$ (median, $\eta < 0.5$) and 0.063 ($\eta \geq 0.5$) that is at most 0.34 %, typically 0.01–0.11 %. Weighting by $(2\pi f)^2$ to suppress the slow mode without a floor was tested and rejected: it moves c_s by +0.02 to +3.3 %, one-signed and largest at high density, which is a spectral-shape systematic and not noise. Taken together the estimator's own systematic is below 0.3 %, against the 1–3 % effects the paper reports.

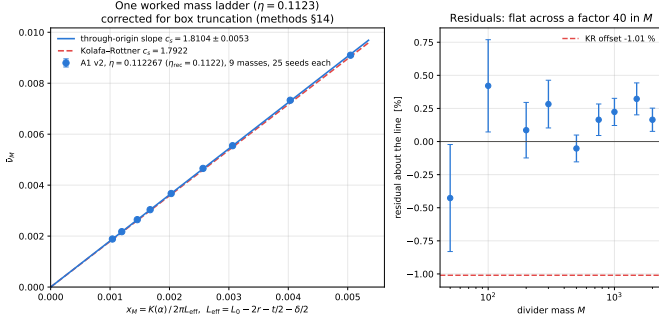


Fig. 4. One worked mass ladder, $\eta = 0.1122$, nine divider masses $M = 50$ –2000, 25 seeds each. **Left:** $\bar{v}_M$ against $x_M = K(\alpha)/2\pi L_{\text{eff}}$ with K the root of $\cot K = \alpha K$; the slope through the origin is c_s . **Right:** residuals about that line, flat across a factor 40 in M . The measured c_s sits 1.0% above Kolafa–Rottner, the finite-size offset discussed in §VI.

IV. THE MASS LADDER

Linearising about the symmetric position makes the divider a mass on two gas springs. Writing the displacement field in each compartment as a standing wave of wavenumber k , imposing continuity at the divider faces and Newton’s law for the divider, $M\ddot{\xi} = -[p_{\text{right}} - p_{\text{left}}]H$, gives with $K = kL$

$$\cot K = \alpha K, \quad \alpha = \frac{M}{2N_s m}, \quad (5)$$

whose lowest root fixes the fundamental,

$$\nu = \frac{c_s K(\alpha)}{2\pi L_{\text{eff}}}. \quad (6)$$

Two limits are worth recording. For a heavy divider $K^2 \rightarrow 2N_s m/M$, so the divider becomes a mass on a spring of stiffness $N_s m c_s^2/L^2$ per compartment — the gas spring is the sound speed. For a light divider $K \rightarrow n\pi/2$ with n odd; the even standing modes are absent because they have a node where the divider sits.

Equation (6) is linear in $K(\alpha)$ through the origin, which makes the mass ladder a self-contained consistency test: a single c_s must place every mass on the same line, with no equation of state involved. We use nine masses, $M = 50$ to 2000, and take c_s as the through-origin slope of $\bar{v}_M$ against $K(\alpha)/2\pi L_{\text{eff}}$, its uncertainty being the scatter of the per-mass values about that line. That scatter, 0.27%, is within a factor two of the bin-rounding floor.

V. RESULTS

Over $\eta \leq 0.4$ the measurement sits +1.01% above Kolafa–Rottner on average at $N = 100$ and never worse than 1.68%. The offset is one-signed and grows slowly with density.

VI. FINITE SIZE

Along $N = 100, 400, 900, 1600$ (and 2500 where available) at fixed η , $c_s(N)$ is fitted with $c_\infty + b/\sqrt{N}$, with b/N and the two-term form bracketing the systematic. At $\eta = 0.02$, $c_\infty = 1.4614 \pm 0.0102$ against 1.4725 from the equation of state (−1.1σ); at $\eta = 0.05$, 1.5699 ± 0.0029 against 1.5666 (+1.1σ). The $N = 100$ offset closes as the box grows: it is finite size, not a defect of the method.

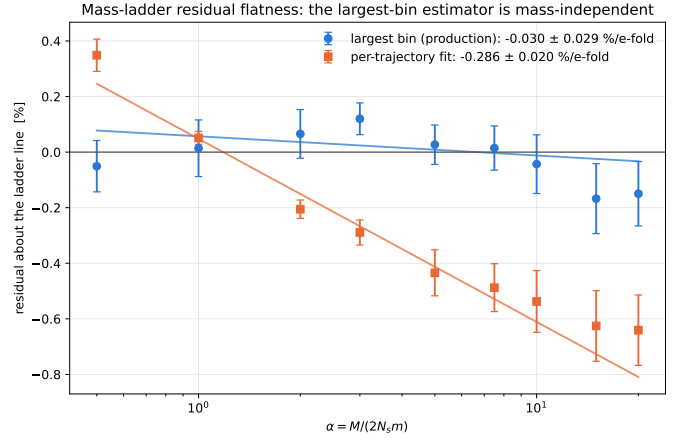


Fig. 5. Mass independence, the ladder’s one assumption, tested. Mean residual about each density’s own ladder line at fixed $\alpha = M/2N_s m$, averaged over the 24 densities with $\eta \leq 0.69$, against $\ln \alpha$. The largest-bin estimator is flat; the per-trajectory fit drifts by an order of magnitude more per e-fold in α , which is why the largest bin is the production estimator.

TABLE I

ROMÁN *et al.* TABLE I RE-MAPPED. η IS RECOMPUTED FROM THEIR L_0 .

L_0	η	c_s (theirs)	dev., $\gamma = 2$	dev., adiabatic
7.5	0.5236	5.99 ± 0.09	+10.6 %	+1.0 %
10	0.3927	3.78 ± 0.08	+5.5 %	+0.9 %
15	0.2618	2.61 ± 0.03	+3.3 %	+1.5 %
20	0.1963	2.20 ± 0.02	+1.9 %	+0.9 %
25	0.1571	2.01 ± 0.02	+1.9 %	+1.3 %
30	0.1309	1.89 ± 0.02	+1.5 %	+1.1 %
35	0.1122	1.81 ± 0.02	+1.3 %	+1.0 %

a) What the offset is not: The pressure measured at the wall of the same finite box exceeds the bulk equation of state by 7–20%, rising with η . Propagated through (1) that would imply a +5 to +16% excess in c_s , five times what is observed. The pressure of a finite box and its sound speed are therefore two *different* finite-size quantities, and the bulk relation between them does not hold at $N = 100$. We report this as a negative result: the wall pressure cannot be used to explain the offset.

VII. COMPARISON WITH ROMÁN *et al.* (2002)

Román *et al.* [3] measured the same observable in the same box ($N_s = 50$, $H = 10$) and report c_s above scaled-particle theory and Henderson by an amount that *grows* with density, from +1.3% at $\eta = 0.112$ to +12.8% at 0.524. That growth is the mapping. They convert Z to c_s with a constant $\gamma = 2$; using (1) on their own numbers gives Table I.

The density-dependent excess becomes a flat +1.10% (spread +0.91 to +1.46%), against our +1.01% at $N = 100$ averaged over $\eta \leq 0.4$. Two independent implementations, 24 years apart, in the same box, with the same offset: that makes the +1% a property of the $N = 100$ divider apparatus rather than of either code. Their own size series (their Table II, $N = 64 \rightarrow 4096$, $c_s = 3.81 \rightarrow 3.75$, extrapolating to 3.74) runs in the same direction as ours. As a transcription check, their scaled-particle and Henderson columns are reproduced to

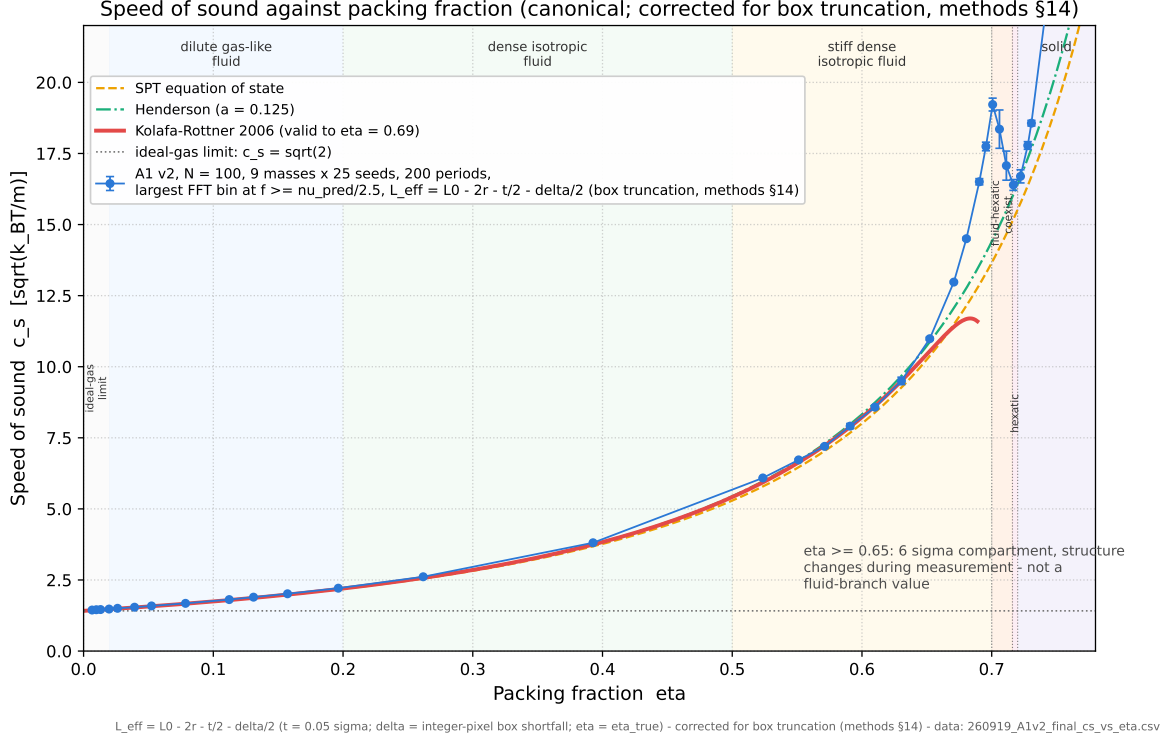


Fig. 6. $c_s(\eta)$ at $N = 100$ against three equations of state, through (1) with no fitted parameter. Shading marks the regimes, with the transition densities taken from Engel *et al.* [2]: liquid–hexatic coexistence over $0.700 \lesssim \eta \lesssim 0.716$ and the hexatic–solid crossover near $\eta = 0.720$. Points beyond $\eta \approx 0.69$ lie in that region, where no fluid reference exists; they are shown but are not counted as a deviation from Kolafa–Rottner.

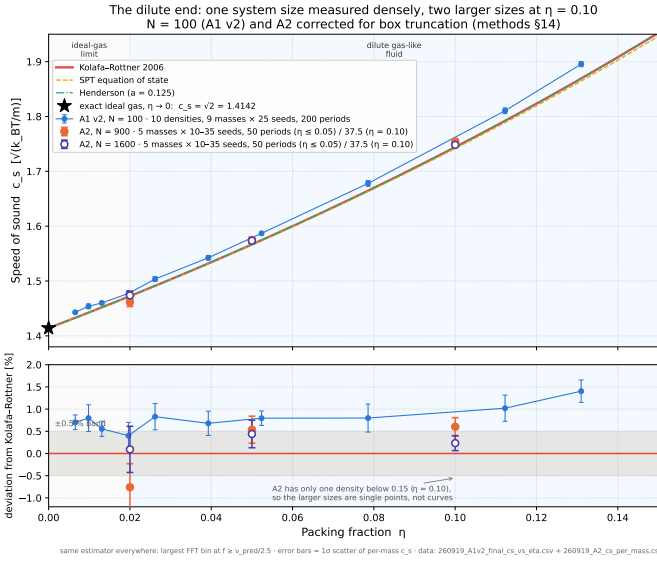


Fig. 7. The dilute end. The $N = 100$ curve sits 0.4–1.4 % above the equation of state; larger systems agree with it within their errors.

every printed digit by our equation-of-state code together with their $\gamma = 2$ mapping.

The two measurements differ at one point, $\eta \approx 0.52$: ours sits +2.6 % and theirs $+1.0 \pm 1.5$ %, which is 1.1σ on their error bar alone, with 100 trajectories against our 25.

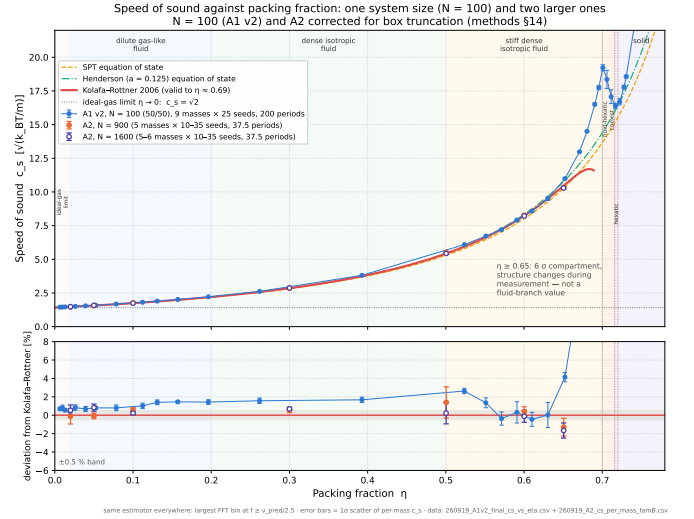


Fig. 8. $N = 100$ against $N = 900$ and 1600 at the same densities.

A. Above $\eta \approx 0.69$: the right comparison is not Kolafa–Rottner

Kolafa–Rottner is a *fluid* equation of state fitted below the transition, so a deviation from it above $\eta \approx 0.69$ measures the extrapolation, not the gas. The reference there is Engel *et al.* [2], who computed the hard-disk equation of state through the melting region with event-chain Monte Carlo, massively parallel Monte Carlo, local Monte Carlo and event-driven molecular dynamics — the same algorithm class used here — and found all four to agree at one tabulated state point,

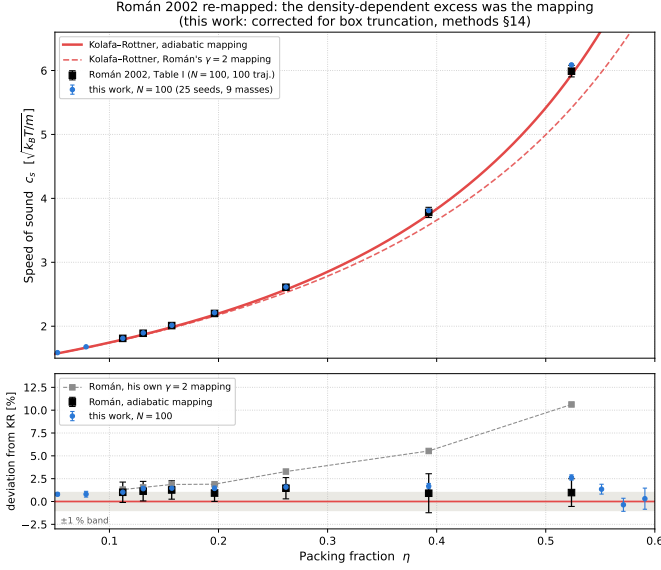


Fig. 9. Román *et al.* re-mapped. Solid red, Kolafa–Rottner through (1); dashed red, the same equation of state through their $\gamma = 2$ mapping; black squares, their Table I with their error bars; blue, this work at $N = 100$. Lower panel: deviation from Kolafa–Rottner.

$\eta = 0.698$, to $\beta P(2\sigma)^2 = 9.1705\text{--}9.1708$ with relative errors $\lesssim 10^{-4}$, i.e. $Z = 10.3187\text{--}10.3191$. That agreement is also the answer to why an event-driven method is used here for an equilibrium quantity at all.

Their equation of state carries a Mayer–Wood loop whose local extrema sit at $\eta \approx 0.702$ and $\eta \approx 0.714$ and whose position is independent of system size. Between those extrema $Z'(\eta) < 0$, and since $c_s^2 = (k_B T/m)(Z + \eta Z' + Z^2)$ a loop in Z must appear as a *dip* in c_s bounded by the same two densities. It does:

	our $c_s(\eta)$, $N = 100$	Engel <i>et al.</i>
local maximum	$\eta = 0.701$	loop extremum ≈ 0.702
local minimum	$\eta = 0.717$	loop extremum ≈ 0.714

both within our grid spacing of 0.005, and the dip is deep: $c_s(0.701) - c_s(0.717) = 2.83$, which is 9.6σ on the error bars plotted in Fig. 10 and 18.4σ on the propagated statistical error alone. **The claim rests on the former**, the conservative of the two: the standard error of the through-origin slope with the per-mass frequency errors propagated from 25 seeds, inflated by $\sqrt{\chi_{\text{red}}^2}$ wherever the nine masses disagree by more than those errors — and in this window they do, $\chi_{\text{red}}^2 = 1.8\text{--}14$. The mass scatter tells the same story independently: 0.45 inside $0.695 \leq \eta \leq 0.720$ against 0.09 outside it, a factor five, which is what a phase-separating and slowly-equilibrating system should do to a resonance measurement.

A caution that applies to the whole window, ours and theirs alike: c_s is built from Z' , and Z' is where every equation of state is weakest. This is a **qualitative** correspondence and is claimed as nothing more. Engel *et al.* work at $N = 256^2$ to 1024^2 ; at $N = 100$ the Mayer–Wood loop is smeared over a width comparable to its own depth, the first-order

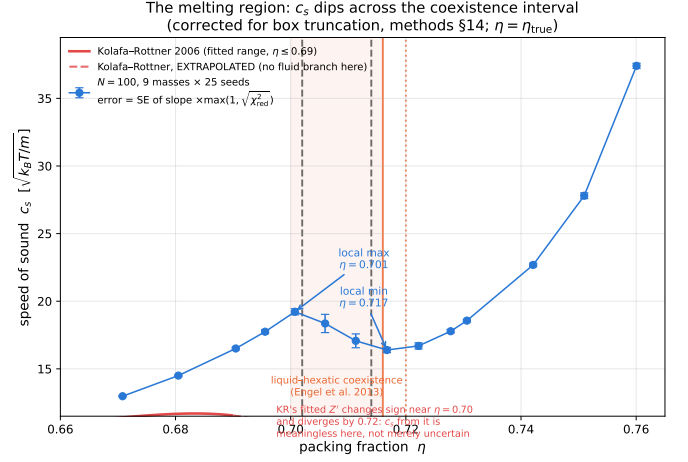


Fig. 10. The melting region. Vertical lines are Engel *et al.*’s independently measured landmarks [2]: the Mayer–Wood loop extrema at $\eta \approx 0.702$ and 0.714 , the end of liquid–hexatic coexistence at 0.716 , and the hexatic–solid crossover at 0.720 . Our c_s turns over inside the same interval. Kolafa–Rottner is solid inside its fitted range and dashed beyond it, and is cut off at $\eta = 0.705$ deliberately: its fitted Z' runs 48 at $\eta = 0.67$, 28 at 0.69 , -9 at 0.700 and $+3386$ at 0.720 , so a c_s built from it there is meaningless rather than merely uncertain. The negative Z' already inside the stated fitted range is also why KR’s own c_s turns over near $\eta = 0.683$.

character cannot be resolved, and their tabulated pressure exists at one density rather than as a table across the range, so no quantitative comparison is available. What the correspondence does establish is that the structure in our $c_s(\eta)$ above 0.69 is the melting transition rather than an estimator artefact.

VIII. OUTLOOK

Two measurements remain. A size ladder at $N \approx 1000$ across the full density range would turn §VI’s two-point extrapolation into a curve; and a size series at $\eta = 0.65$ would test whether the offset continues to close where the compartment is only six disk diameters wide. Both are cluster jobs. Above $\eta \approx 0.69$ the comparison is with Engel *et al.* [2] rather than with a fluid equation of state (§VII-A); turning the qualitative correspondence found there into a quantitative one needs a global equation of state with hexatic and solid branches [4] and a box large enough to resolve the Mayer–Wood loop, neither of which is available at $N = 100$.

REPRODUCIBILITY

Every campaign directory carries the exact command and the estimator geometry line. The geometry constant and the estimator share one definition, and the analysis refuses to run on data whose recorded wall thickness disagrees with it.

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